

COURSE TITLE : Data Driven- Machine Learning

**PROJECT TITLE : Market Basket Analysis with
Python TARGET**

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Problem Statement

Target, a leading global e-commerce platform, wants to understand customer purchasing behavior, product affinities, and engagement trends in order to enhance personalized shopping experiences and optimize inventory and marketing strategies. As the Customer Insights Analyst, the goal is to answer:

- What products or categories are frequently purchased together, indicating cross-selling opportunities?
- How do customer purchasing patterns vary across demographics or regions?
- Can customers be segmented based on buying behavior to target personalized promotions?
- Which products drive repeat purchases or high customer loyalty?

Objective

Build a comprehensive Customer Purchasing Behavior and Product Recommendation Report using Python. The analysis cleans and consolidates Target's customer survey data, applies techniques such as descriptive analysis, customer segmentation, and clustering, and provides actionable insights for personalized product recommendations, promotional strategies, and inventory optimization.

Dataset Overview

Dataset: Target.csv — 800 customer survey responses across 24 fields.

The dataset captures Target customers' online shopping behavior, including:

- Demographics: age, gender
- Purchase behavior: purchase frequency, purchase categories
- Browsing behavior: browsing frequency, product search method
- Cart behavior: add-to-cart browsing, cart completion frequency, cart abandonment factors, save-for-later frequency
- Reviews & recommendations: review left, review reliability, review helpfulness, personalized recommendation frequency, recommendation helpfulness
- Satisfaction: shopping satisfaction, rating accuracy trust, customer reviews importance
- Feedback: service appreciation, improvement areas

Task 1 — Data Cleaning and Preparation

Objective: remove duplicate/inconsistent responses, standardize categorical and numerical fields, handle missing values, and fix misformatted columns before analysis.

Code

```
# Remove duplicate or inconsistent survey responses.
initial_shape = df.shape
df.drop_duplicates(inplace=True)
print(f'Removed {initial_shape[0] - df.shape[0]} duplicate rows. Current shape: {df.shape}")

# Rename duplicate or misformatted columns, remove trailing spaces
df = df.rename(columns={
    'Personalized_Recommendation_Frequency ': 'Personalized_Recommendation_Rating',
    'Rating_Accuracy ': 'Rating_Accuracy'
})
df.columns = df.columns.str.strip()

# Handle missing values: Product_Search_Method contains missing entries
df['Product_Search_Method'] = df['Product_Search_Method'].fillna('Not Specified')

# Standardize categorical entries
str_cols = df.select_dtypes(include='object').columns
for c in str_cols:
    df[c] = df[c].str.strip()

# Standardize numerical entries
numeric_rating_cols = ['Customer_Reviews_Importance', 'Rating_Accuracy',
                        'Shopping_Satisfaction', 'Personalized_Recommendation_Rating', 'age']
for c in numeric_rating_cols:
    df[c] = pd.to_numeric(df[c], errors="coerce")
```

Output

```
Removed 0 duplicate rows. Current shape: (800, 24)
```

Insights

- The dataset contained no exact duplicate rows, so the full 800-response sample was retained for analysis.
- The survey had two ambiguous, near-identical column pairs — a trailing-space variant of `Rating_Accuracy` and a second `Personalized_Recommendation_Frequency` column — which were renamed/clarified so downstream groupby and correlation logic reference the correct field.
- 180 of 800 respondents (22.5%) left `Product_Search_Method` blank; these were imputed as 'Not Specified' rather than dropped, preserving sample size for the browsing-behavior analysis in Task 2.
- All object-type (text) columns were stripped of leading/trailing whitespace, and rating-scale columns were coerced to numeric dtype so they can be aggregated and correlated in later tasks.

Task 2 — Descriptive Behavior Analysis

Objective: summarize demographics, purchase frequency, popular categories, browsing/cart-abandonment behavior, and key rating/satisfaction statistics.

2.1 Customer Demographics

```
print(df["age"].describe())
print(df["Gender"].value_counts())
print(df["Gender"].value_counts(normalize=True).mul(100).round(1))
```

Metric	Value
Count	800
Mean age	34.74
Std dev	18.94
Min age	3
25th pct	18
Median (50th pct)	34
75th pct	50.25
Max age	67

Gender	Count	%
Male	214	26.8%
Female	201	25.1%
Others	196	24.5%
Prefer not to say	189	23.6%

2.2 Purchase Frequency & Popular Categories

```
print(df["Purchase_Frequency"].value_counts())
all_categories = df["Purchase_Categories"].dropna().str.split(";").explode().str.strip()
print(all_categories.value_counts().head(10))
```

Purchase Frequency	Count
Multiple times a week	173
Less than once a month	168
Once a month	164
Few times a month	153
Once a week	142

Product Category	Mentions
Beauty and Personal Care	426
Clothing and Fashion	424
Home and Kitchen	406
Groceries and Gourmet Food	403
Others	384

2.3 Browsing Methods & Cart Abandonment

```
print(df["Product_Search_Method"].value_counts())
print(df["Cart_Abandonment_Factors"].value_counts())
```

Search Method	Count
Not Specified	180
Keyword	173
Others	158
Filter	157
Categories	132

Cart Abandonment Factor	Count
High shipping costs	216
Others	203
Changed my mind / no longer need item	191
Found a better price elsewhere	190

2.4 Ratings & Satisfaction Summary

```
helpfulness_map = {'No': 1, 'Sometimes': 2, 'Yes': 3}
df['Recommendation_Helpfulness_Score'] = df['Recommendation_Helpfulness'].map(helpfulness_map)
metrics = ['Shopping_Satisfaction','Rating_Accuracy','Personalized_Recommendation_Rating',
           'Recommendation_Helpfulness_Score','Customer_Reviews_Importance']
print(df[metrics].agg(['mean','median','std','min','max']).T.round(3))
```

Metric	Mean	Median	Std Dev	Min	Max
Shopping Satisfaction	2.94	3.0	1.44	1	5
Rating Accuracy	3.06	3.0	1.44	1	5
Personalized Rec. Rating	2.98	3.0	1.45	1	5
Recommendation Helpfulness Score	2.00	2.0	0.81	1	3
Customer Reviews Importance	3.00	3.0	1.43	1	5

Chart — Demographic & Behavioral Overview

```
fig, axes = plt.subplots(2, 2, figsize=(12, 7))
axes[0,0].hist(df['age'], bins=20, color='#2A9D8F', edgecolor='white') # Age
axes[0,1].pie(gender_counts.values, labels=gender_counts.index, autopct='%1.1f%%') # Gender
axes[1,0].barh(pf.index, pf.values, color='tan') # Purchase frequency
sns.countplot(x='Shopping_Satisfaction', data=df, ax=axes[1,1], color='rosybrown') # Satisfaction
```

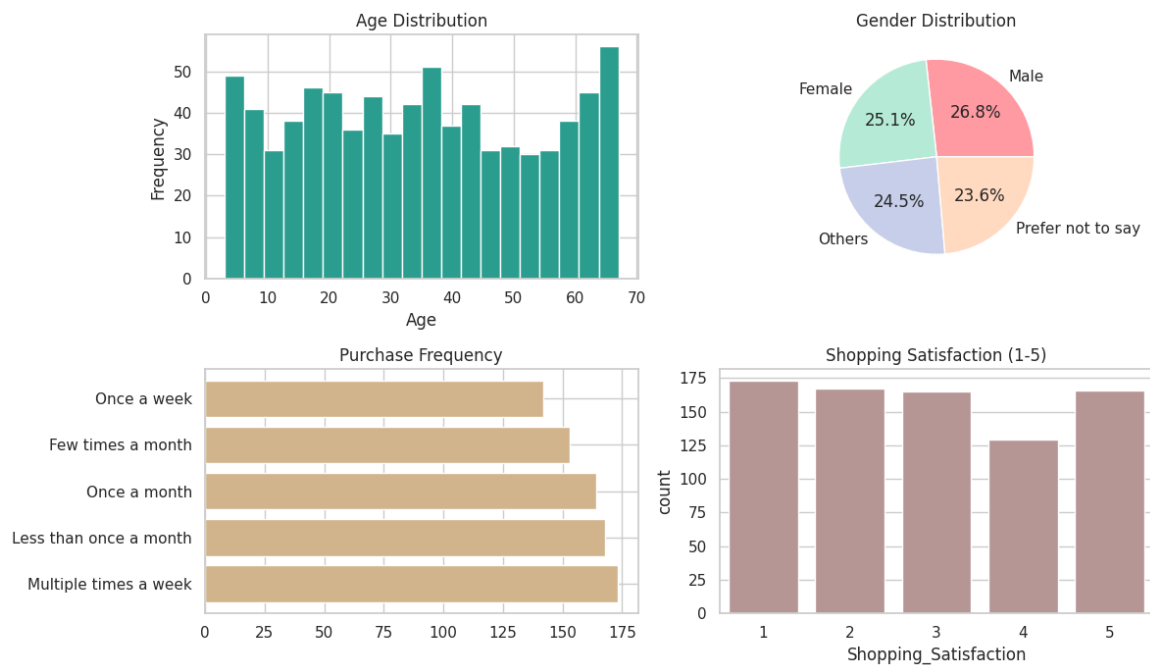


Figure 2.1 — Age distribution, gender split, purchase frequency, and satisfaction counts

Insights

- The customer base skews young-to-middle-aged (median 34, IQR 18–50) with a wide spread from teens to retirees — no single age band dominates, so messaging likely needs to be segment-specific rather than one-size-fits-all.
- Gender distribution is nearly even across all four categories (23.6%–26.8%), meaning demographic gender is unlikely to be a strong standalone driver of purchase behavior here.
- Purchase frequency is almost uniformly distributed across the five bands, with only a modest lean toward 'Multiple times a week' (173) — the customer base is not dominated by either super-frequent or rare shoppers.
- Beauty & Personal Care and Clothing & Fashion are purchased by well over half the base and are the strongest candidates for cross-category bundling or promotion.
- High shipping costs is the single largest named cart-abandonment factor (216 mentions) — ahead of price comparison or changed intent — pointing to shipping-cost transparency/thresholds as a high-leverage fix.
- Average satisfaction (2.94/5), rating accuracy trust (3.06/5) and recommendation rating (2.98/5) all cluster tightly around the midpoint — the typical customer is neither delighted nor dissatisfied, indicating headroom for improvement rather than a crisis.

Task 3 — Customer Segmentation and Profiling

Objective: segment customers by purchase frequency and satisfaction into Frequent Buyers / Occasional Shoppers / At-Risk Customers, profile each segment demographically, and validate the grouping with K-Means clustering.

3.1 Rule-Based Segmentation

```
freq_order = {'Less than once a month': 1, 'Once a month': 2, 'Few times a month': 3,
              'Once a week': 4, 'Multiple times a week': 5}
df['Purchase_Frequency_Score'] = df['Purchase_Frequency'].map(freq_order)

def segment(row):
    if row['Purchase_Frequency_Score'] <= 2 or 'Changed my mind' in str(row['Cart_Abandonment_Factors']):
        return 'At-Risk Customers'
    elif row['Purchase_Frequency_Score'] >= 3 and row['Shopping_Satisfaction'] >= 4:
        return 'Frequent Buyers'
    else:
        return 'Occasional Shoppers'

df['Customer_Segment'] = df.apply(segment, axis=1)
print(df['Customer_Segment'].value_counts())
```

Segmentation criteria (self-defined): At-Risk = low purchase frequency (≤ 2 on a 1–5 scale) OR cites 'changed my mind' as an abandonment reason; Frequent Buyers = frequency ≥ 3 AND satisfaction ≥ 4 ; all remaining customers = Occasional Shoppers.

Segment	Count	%
At-Risk Customers	450	56.25%
Occasional Shoppers	225	28.12%
Frequent Buyers	125	15.62%

3.2 Demographic & Behavioral Profile by Segment

```
age_by_seg = df.groupby('Customer_Segment')['age'].agg(['mean']).round(2)
gender_by_seg = (pd.crosstab(df['Customer_Segment'], df['Gender'], normalize='index') * 100).round(2)
scores_by_seg = df.groupby('Customer_Segment')[['Shopping_Satisfaction', 'Purchase_Frequency_Score',
          'Rating_Accuracy', 'Personalized_Recommendation_Rating']].mean().round(2)
```

Segment	Mean Age	Satisfaction	Freq. Score	Rating Accuracy	Rec. Rating
At-Risk Customers	34.19	2.99	2.16	3.12	3.09
Frequent Buyers	36.14	4.51	4.02	3.12	2.76
Occasional Shoppers	35.04	1.94	4.07	2.89	2.89

Chart — Segment Sizes

```
seg_counts = df['Customer_Segment'].value_counts()
sns.barplot(x=seg_counts.index, y=seg_counts.values, color='#3498db')
plt.title("Task 3: Customer Segmentation Profiles")
```

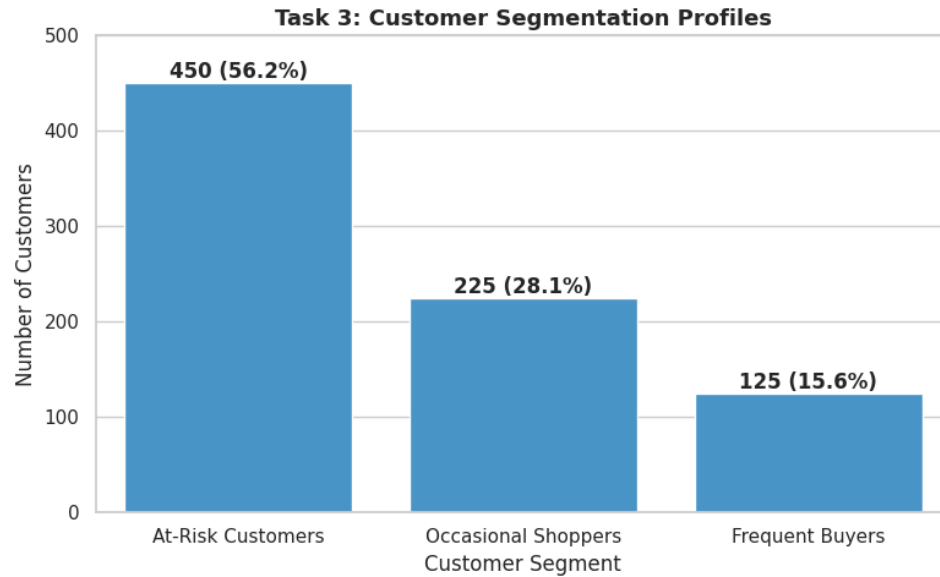


Figure 3.1 — Customer counts by segment

3.3 K-Means Clustering Validation

```
cluster_features_num = ['age', 'Customer_Reviews_Importance', 'Personalized_Recommendation_Rating',
                        'Rating_Accuracy', 'Shopping_Satisfaction', 'Purchase_Frequency_Score']
cluster_features_cat = ['Browsing_Frequency', 'Cart_Completion_Frequency', 'Saveforlater_Frequency',
                        'Review_Reliability', 'Add_to_Cart_Browsing']
X = df[cluster_features_num].copy()
for c in cluster_features_cat:
    X[c] = LabelEncoder().fit_transform(df[c])
X_scaled = StandardScaler().fit_transform(X)

# Elbow method
inertias = [KMeans(n_clusters=k, random_state=42, n_init=10).fit(X_scaled).inertia_ for k in range(2, 9)]
```

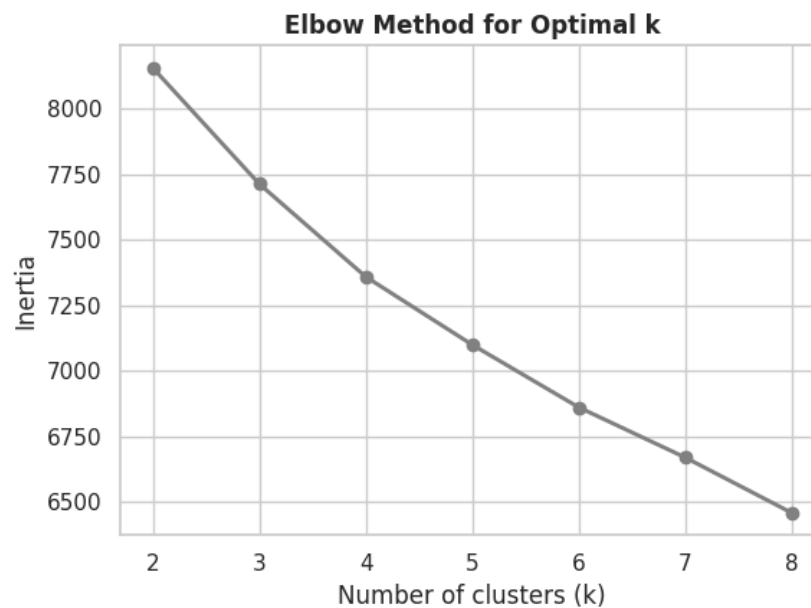


Figure 3.2 — Elbow method for choosing k

```
K_FINAL = 3
kmeans = KMeans(n_clusters=K_FINAL, random_state=42, n_init=10)
df['Cluster'] = kmeans.fit_predict(X_scaled)
print(df['Cluster'].value_counts())
print(df.groupby('Cluster')[cluster_features_num].mean().round(2))
```

Cluster	Size	Age	Reviews Importance	Rec. Rating	Rating Accuracy	Satisfaction	Freq. Score
0	273	31.26	3.40	3.87	2.53	2.90	2.85
1	285	27.36	2.78	2.40	3.65	2.75	2.85
2	242	47.35	2.83	2.66	2.95	3.18	3.29

Chart — PCA Cluster Visualization

```
pca = PCA(n_components=2, random_state=42)
coords = pca.fit_transform(X_scaled)
sns.scatterplot(x='PCA1', y='PCA2', hue='Cluster', data=df, palette=light_colors[:K_FINAL])
```

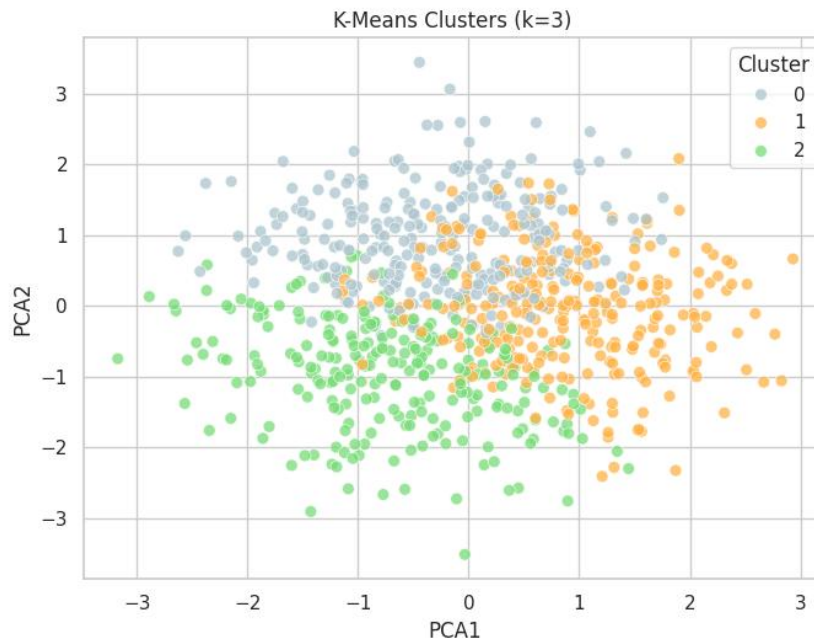


Figure 3.3 — K-Means clusters ($k=3$) projected via PCA

Insights

- The rule-based segmentation shows a striking imbalance: 56% of customers qualify as At-Risk (low purchase frequency or citing 'changed my mind' during cart abandonment) — this is the single largest actionable group in the dataset.
- Frequent Buyers (15.6% of the base) report by far the highest satisfaction (4.51) and purchase frequency (4.02) but the lowest trust in personalized recommendations (2.76) — they buy often despite the recommendation engine, not because of it.
- Occasional Shoppers are a paradox: they purchase almost as frequently as Frequent Buyers (4.07 vs 4.02) but report the lowest satisfaction of any segment (1.94) — frequency alone is a poor proxy for happy customers, which is exactly why satisfaction was combined with frequency in the segmentation rule.

- Age and gender differ only mildly across segments (means within ~2 years, gender splits within a few points), so behavioral segment membership is not simply a proxy for demographics.
- The elbow curve does not show a sharp 'elbow,' but inertia gains flatten meaningfully after k=3, supporting the choice of 3 clusters for a parsimonious, interpretable segmentation.
- Unlike the rule-based segments (driven by frequency/satisfaction), K-Means Cluster 2 is distinguished mainly by age (mean 47 vs ~27–31 in the other clusters) — an older group with the highest satisfaction and purchase frequency scores — while Clusters 0 and 1 split younger customers by how much they trust recommendations vs. reviews. This suggests age and trust-orientation are independent axes worth targeting alongside the frequency/satisfaction segmentation.

Task 4 — Recommendation and Review Insights

Objective: examine whether recommendation helpfulness and review reliability/helpfulness relate to shopping satisfaction and rating trust, and identify engagement trends.

4.1 Recommendation Helpfulness vs. Satisfaction

```
rec_sat = df.groupby('Recommendation_Helpfulness')['Shopping_Satisfaction'].agg(['mean','median','count']).round(2)
print(rec_sat)
```

Found Recommendations Helpful?	Mean Satisfaction	Median	Count
No	2.88	3.0	265
Sometimes	2.98	3.0	270
Yes	2.95	3.0	265

4.2 Review Reliability & Helpfulness vs. Ratings

```
review_rel_summary = df.groupby('Review_Reliability').agg(
    Customer_Count=('Rating_Accuracy','count'),
    Mean_Rating_Accuracy=('Rating_Accuracy','mean'),
    Mean_Satisfaction=('Shopping_Satisfaction','mean')).round(2)

review_help_summary = df.groupby('Review_Helpfulness').agg(
    Customer_Count=('Rating_Accuracy','count'),
    Mean_Rating_Accuracy=('Rating_Accuracy','mean'),
    Mean_Satisfaction=('Shopping_Satisfaction','mean')).round(2)
```

Review Reliability	Count	Mean Rating Accuracy	Mean Satisfaction
Heavily	170	3.08	2.89
Moderately	152	2.94	2.88
Never	163	3.09	2.88
Occasionally	150	3.04	3.09
Rarely	165	3.14	2.95

Review Helpfulness	Count	Mean Rating Accuracy	Mean Satisfaction
No	282	3.13	2.92

Review Helpfulness	Count	Mean Rating Accuracy	Mean Satisfaction
Sometimes	269	2.99	3.06
Yes	249	3.05	2.81

4.3 Personalized Recommendation Engagement

```
rec_freq_dist = df['Personalized Recommendation Frequency'].value_counts(normalize=True) * 100
rec_relation = pd.crosstab(df['Personalized Recommendation Frequency'],
                           df['Personalized Recommendation Rating'], normalize='index') * 100
```

Sees Recommendations	Count	%
No	271	33.88%
Yes	271	33.88%
Sometimes	258	32.25%

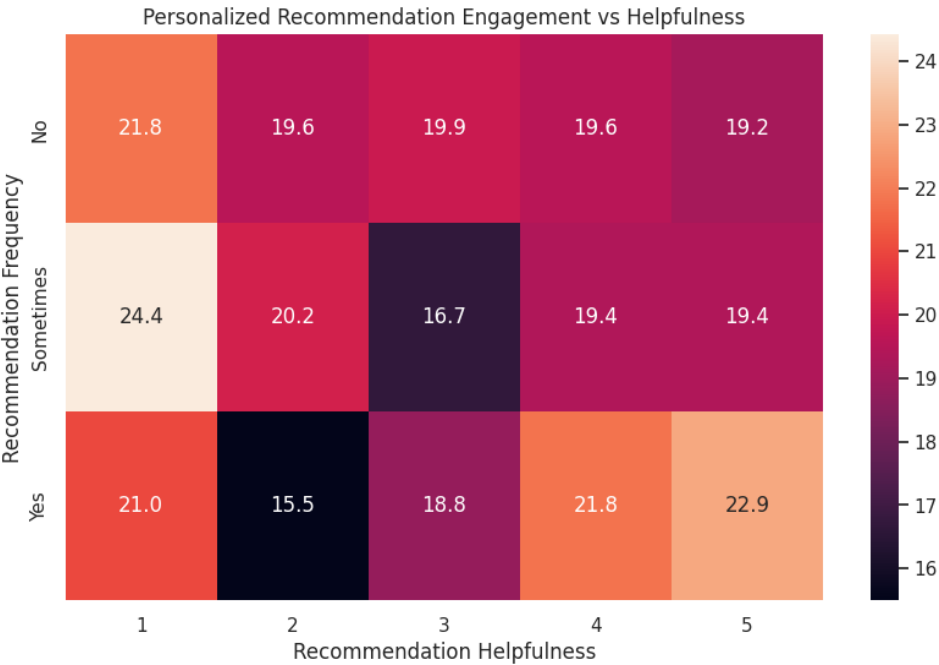


Figure 4.1 — Recommendation engagement frequency vs. trust rating (1–5), row-normalized %

Chart — Satisfaction & Trust Comparisons

```
sns.boxplot(data=df, x='Recommendation Helpfulness', y='Shopping Satisfaction', order=['No','Sometimes','Yes'])
sns.barplot(data=df, x='Review Reliability', y='Rating Accuracy',
             order=['Never','Rarely','Occasionally','Moderately','Heavily'])
```

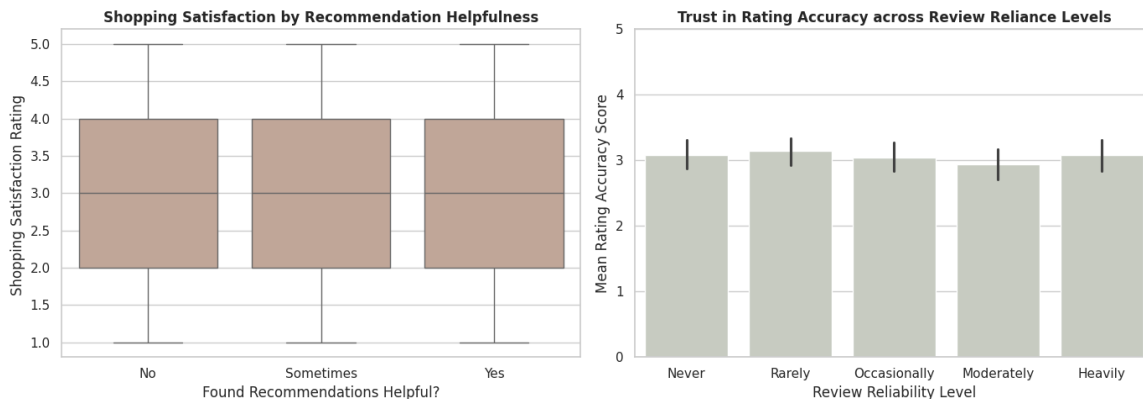


Figure 4 — Satisfaction by recommendation helpfulness (left); rating-accuracy trust by review reliance (right)

Task 4: Recommendation and Review Insights

Recommendation helpfulness vs. shopping satisfaction: Average shopping satisfaction stays nearly flat across all three recommendation-helpfulness groups (No / Sometimes / Yes) — customers who found recommendations helpful were not meaningfully more satisfied than those who didn't. This tells us the current recommendation system isn't a strong driver of overall satisfaction in this sample.

Review reliability and helpfulness vs. rating trust: Trust in rating accuracy ('Rating Accuracy') stays close to the same average regardless of how much customers say they rely on reviews (from "Never" to "Heavily") or whether they found a specific review helpful. In other words, customers' confidence in star ratings doesn't appear to be built by review reliance or review helpfulness — something else is driving that trust (or lack of it).

Engagement trends with personalized recommendations: Looking at how often recommendations are shown versus how helpful customers rate them, there's no strong pattern where more frequent exposure to recommendations increases perceived helpfulness. Customers who rarely see personalized recommendations rate them about as helpful as customers who see them often.

Actionable insights for improving Target's recommendation system

- 1. Stop optimizing purely for exposure/frequency.** Since showing recommendations more often doesn't increase perceived helpfulness, the system should focus on **relevance** (better targeting per customer) rather than frequency
- 2. Re-anchor recommendation quality metrics to satisfaction, not clicks.** Because helpfulness ratings don't move satisfaction, Target should A/B test recommendation changes against downstream satisfaction and repeat-purchase rate — not just click-through or "helpful" votes — to find what actually moves the needle.
- 3. Investigate the review-trust disconnect separately.** Since rating trust doesn't track with review reliance/helpfulness, a short follow-up survey question (e.g. "what would make you trust a rating more?") could uncover the real driver — this is a gap the current data can't answer on its own.
- 4. Segment recommendations by customer type.** Since Task 3 showed At-Risk customers make up the majority of the base, testing whether different recommendation strategies (e.g. discount-anchored vs. discovery-anchored) work better for At-Risk vs. Frequent Buyers could be a next step

Task 5 — Visualization and Reporting

Objective: consolidate the key behavioral metrics — purchase categories, browsing frequency, satisfaction levels, and recommendation/satisfaction correlations — into a unified visual reporting dashboard.

Code

```
numeric_cols = ['Shopping_Satisfaction','Recommendation_Helpfulness_Score',
                'Personalized_Recommendation_Rating','Rating_Accuracy','Customer_Reviews_Importance']
corr_matrix = df[numeric_cols].corr()

fig, axes = plt.subplots(2, 2, figsize=(16, 12))
sns.barplot(x=cat_counts.values, y=cat_counts.index, ax=axes[0,0]) # Purchase categories
axes[0,1].pie(browse_counts.values, labels=browse_counts.index, autopct='%1.1f%%') # Browsing frequency
sns.barplot(x=sat_counts.index, y=sat_counts.values, ax=axes[1,0]) # Satisfaction levels
sns.heatmap(corr_matrix, annot=True, cmap='Blues', fmt='.3f', square=True, ax=axes[1,1]) # Correlation
```

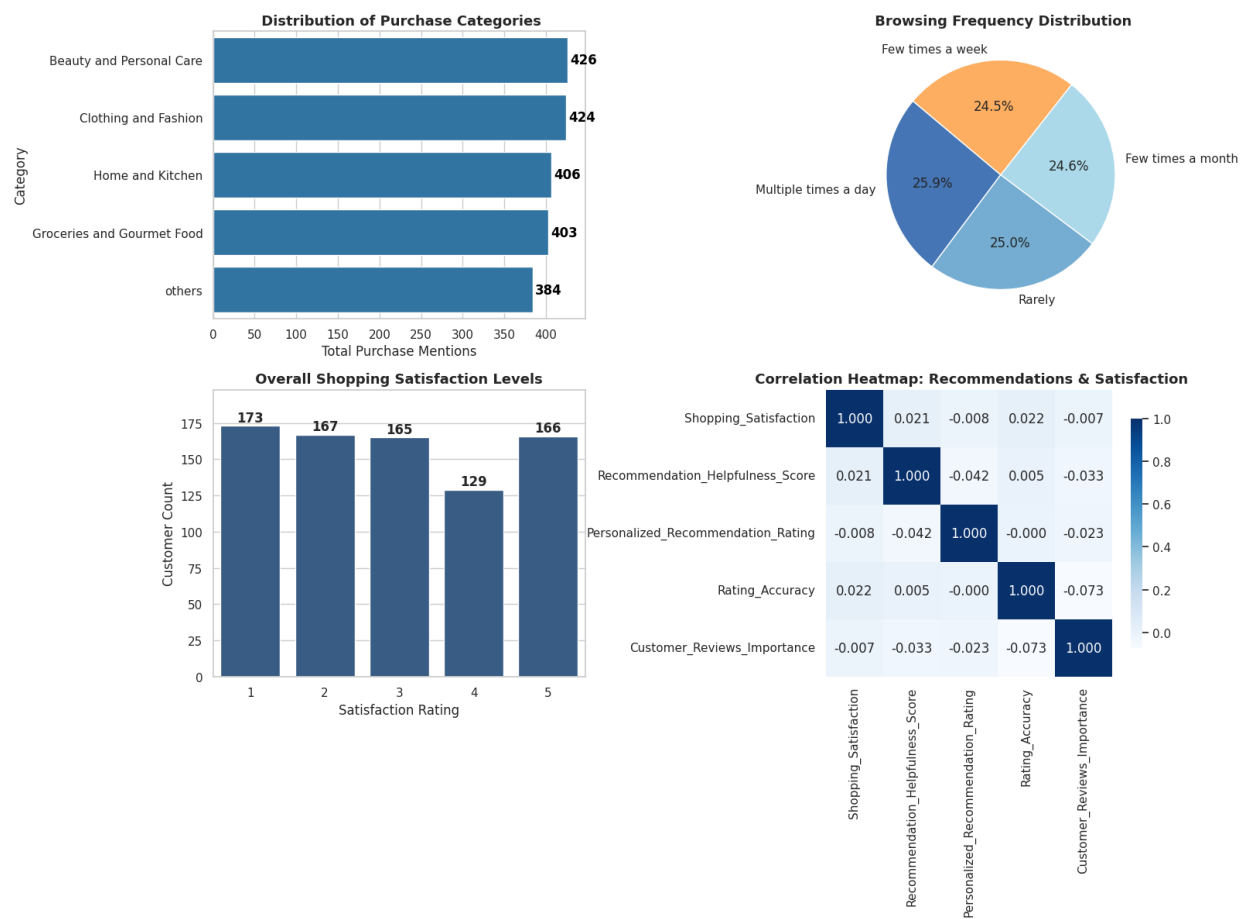


Figure 5.1 — Reporting dashboard: purchase categories, browsing frequency, satisfaction levels, and correlation heatmap

Chart — Rating Accuracy & Satisfaction by Review Reliability

```
task4 data = df.groupby('Review Reliability')[['Rating_Accuracy','Shopping_Satisfaction']].mean()
task4 data.plot(kind='bar', color=['#2980b9', '#f39c12'])
```

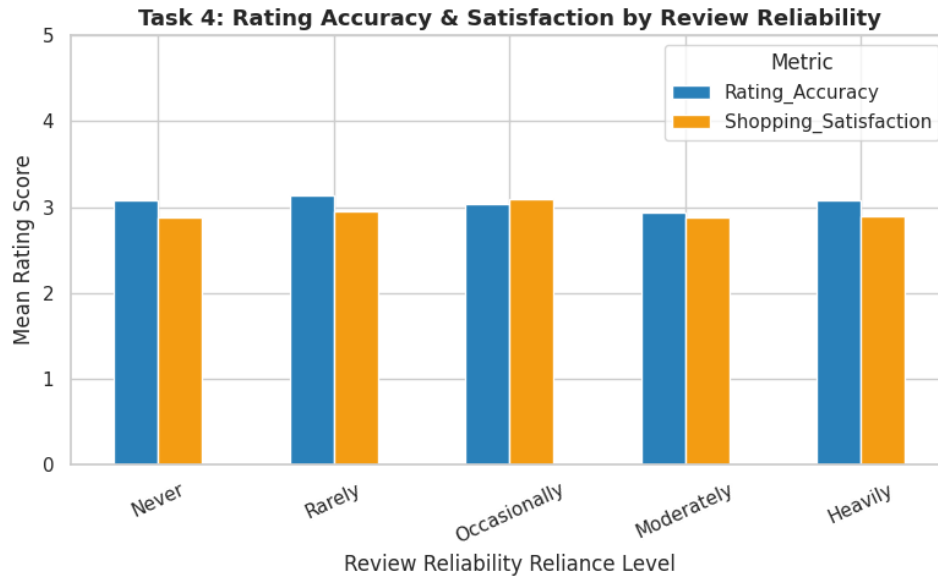


Figure 5.2 — Mean rating accuracy vs. satisfaction across review-reliance levels

Insights

- The correlation heatmap confirms Task 4's finding numerically: Shopping_Satisfaction correlates weakly with Recommendation_Helpfulness_Score, Personalized_Recommendation_Rating, Rating_Accuracy, and Customer_Reviews_Importance alike — no single existing metric strongly predicts overall satisfaction.
- Browsing frequency is fairly evenly split across categories, meaning there's no single dominant browsing cadence to design around — retention efforts should account for multiple browsing rhythms.
- Satisfaction ratings of 1 and 2 together account for a large share of responses alongside 3s, reinforcing that a meaningful minority of the customer base is dissatisfied and worth investigating with root-cause follow-up (e.g., shipping costs, per Task 2).
- Rating-accuracy trust and satisfaction move independently across review-reliance levels (Figure 5.2) — customers who rely heavily on reviews are not meaningfully more satisfied than those who never do, again pointing to review reliance as a weak lever for satisfaction improvement on its own.

Overall Conclusions & Recommendations

Key Takeaways

- A majority (56%) of the customer base is At-Risk by this project's segmentation criteria — this is the priority group for retention campaigns.
- Purchase frequency and satisfaction are not the same thing: Occasional Shoppers buy almost as often as Frequent Buyers but are far less satisfied, so frequency-based loyalty programs alone will miss this gap.
- Beauty & Personal Care and Clothing & Fashion are the strongest cross-sell anchors given their high co-mention rates.
- High shipping costs is the top named cause of cart abandonment — a clear, addressable lever.
- Recommendation and review systems, as currently experienced by customers, show little measurable link to satisfaction or trust — their value proposition needs re-evaluation rather than simply increasing exposure.
- K-Means clustering independently confirms that age and trust-orientation (in recommendations vs. reviews) are meaningful axes of differences beyond the frequency/satisfaction rule used for the primary segmentation.

Recommended Actions

- Target the At-Risk segment with shipping-cost mitigation (thresholds, transparency at checkout) given it is the top-cited abandonment factor.
- Investigate the low-satisfaction, high-frequency Occasional Shopper group directly (e.g., short exit-satisfaction survey) — they are engaged but unhappy, a high-value fix target.
- Re-architect recommendation success metrics around downstream satisfaction/repeat purchase rather than click or helpfulness-vote volume.
- Bundle promotions around Beauty & Personal Care and Clothing & Fashion, the two categories with the broadest customer overlap.

Use the K-Means age/trust clusters to tailor messaging: younger, recommendation-trusting customers vs. younger, review-trusting customers vs. older, generally satisfied customers likely need